

常微分方程 在一个微分方程中,自变量的个数为一个

偏微分方程: 在一个微分方程中,自变量的个数为两个或以上

1. **微分方程的阶:** 微分方程中出现的未知函数的最高阶导数或微分

2. **n阶线性微分方程的一般形式** $\frac{d^ny}{dx^n} + a_1 \frac{d^{n-1}y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = f(x)$

3. **可分离变量的微分方程** —— **分离变量法**

形式: $\frac{dy}{dx} = f(x)g(y) \quad (g(y) \neq 0)$

$$\frac{1}{g(y)} dy = f(x) dx$$

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

$$G(y) = F(x) + C$$

注:可能会丢失 $G(y)=0$ 的特解.

Eg: $\frac{dy}{dx} = 3x^2y$ 的通解

$$\frac{dy}{y} = 3x^2 dx$$

$$\int \frac{dy}{y} = \int 3x^2 dx$$

$$\ln|y| = x^3 + C_1$$

$$|y| = e^{x^3 + C_1} = e^{C_1} e^{x^3}$$

$$y = \pm e^{C_1} e^{x^3}$$

$$y = C e^{x^3} \quad (C \in \mathbb{R})$$

4. **可化为可分离变量方程**

① **齐次方程**

$$\frac{dy}{dx} = \varphi\left(\frac{y}{x}\right)$$

$$\text{令 } u = \frac{y}{x} \text{ 则 } y = ux$$

$$\Rightarrow u + x \frac{du}{dx} = \varphi(u)$$

$$\Rightarrow \frac{du}{\varphi(u)-u} = \frac{dx}{x}$$

② **利用变量代换求微分方程的通解**

Eg: 求 $\frac{dy}{dx} = 1+x+y^2$ 的通解

$$\text{令 } u = x+y \text{ 则 } \frac{du}{dx} - 1 = u^2$$

$$\frac{du}{dx} = u^2 + 1$$

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Eg: $\frac{dy}{dx} = \frac{x+y+4}{x-y-6}$

先求 $\begin{cases} x+y+4=0 \\ x-y-6=0 \end{cases}$

可得: $x=1, y=-5$

令 $X=x-1, Y=y+5$

$$\frac{dY}{dX} = \frac{X+Y}{X-Y} = \frac{1+\frac{Y}{X}}{1-\frac{Y}{X}}$$

$$\text{令 } u = \frac{Y}{X} \text{ 则 } \frac{dY}{dX} = X \frac{du}{dX} + u = \frac{1+u}{1-u}$$

$$\Rightarrow \frac{dY}{dX} \cdot X = \frac{1+u}{1-u} - u = \frac{1-u^2}{1-u}$$

$$\Rightarrow \frac{1-u}{u^2+1} du = \frac{dX}{X}$$

$$\Rightarrow \arctan u - \frac{1}{2} \ln|u^2+1| = \ln|X| + C$$

$$\text{将 } u = \frac{Y}{X}, X=x-1, Y=y+5 \text{ 代入}$$

一阶线性分方程标准形式: $\frac{dy}{dx} + p(x)y = q(x) \rightarrow$ 自变量不含 y

若 $q(x)=0$ 则称为齐次方程 $\rightarrow$ 自由项为 0

1. 解齐次方程: $\frac{dy}{dx} + p(x)y = 0$

$$\int \frac{dy}{y} = \int -p(x) dx$$

$$\ln|y| = \int -p(x) dx + \ln|C|$$

$$|y| = |C| e^{\int -p(x) dx}$$

$$y = C e^{\int -p(x) dx}$$

积分因子法: $e^{\int p(x) dx}$

$$\frac{dy}{dx} \cdot e^{\int p(x) dx} + p(x)y e^{\int p(x) dx} = q(x) e^{\int p(x) dx}$$

$$(y e^{\int p(x) dx})' = q(x) e^{\int p(x) dx}$$

$$y e^{\int p(x) dx} = \int q(x) e^{\int p(x) dx} dx + C$$

$$y = e^{-\int p(x) dx} \left[\int q(x) e^{\int p(x) dx} dx + C \right]$$

2. 解非齐次方程: $\frac{dy}{dx} + p(x)y = q(x)$

常数变易法

$$\frac{dy}{dx} = y \left(\frac{dy}{y} - p(x) \right)$$

$$\int \frac{dy}{y} = \int \left(\frac{q(x)}{y} - p(x) \right) dx$$

$$\ln|y| = - \int p(x) dx + \int \frac{q(x)}{y} dx + C$$

$$y = u(x) e^{-\int p(x) dx}$$

$$y = e^{-\int p(x) dx} \cdot \left[\int e^{\int p(x) dx} q(x) dx + C \right]$$

$$= C e^{-\int p(x) dx} + e^{-\int p(x) dx} \int e^{\int p(x) dx} q(x) dx$$

可化为一阶线性方程:

Bernoulli: (伯努利) 方程: $\frac{dy}{dx} + p(x)y = q(x)y^n, n \neq 0, 1$

$$y^{-n} \frac{dy}{dx} + p(x)y^{1-n} = q(x) \quad (n \neq 0, 1)$$

$$\text{令 } z = y^{1-n}, \quad \frac{dz}{dx} = \frac{d(y^{1-n})}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

$$\frac{dy}{dx} \cdot y^{-n} = \frac{1}{1-n} \cdot \frac{dz}{dx}$$

$$\therefore \frac{dz}{dx} \cdot \frac{1}{1-n} + p(x) \cdot z = q(x)$$

$$\therefore \frac{dz}{dx} + (1-n)p(x)z = (1-n)q(x)$$

$$\text{Eq: } 2y y' + 2xy^2 = x e^{-x^2}$$

$$y' + xy = \frac{x e^{-x^2}}{2y}$$

$$\text{令 } z = y^2, \quad \frac{dz}{dx} \cdot \frac{1}{2} + xz = \frac{x e^{-x^2}}{2}$$

$$\frac{dz}{dx} + 2xz = x e^{-x^2}$$

$$y^2 = z = e^{-\int 2x dx} \left(C + \int x e^{-x^2} \cdot e^{\int 2x dx} dx \right)$$

$$= e^{-x^2} \left(C + \int x e^{-x^2} \cdot e^{x^2} dx \right)$$

$$= e^{-x^2} \left(C + \frac{1}{2} x^2 \right)$$

$$\text{Eq: } \frac{dy}{dx} = \frac{1}{x+y} \quad \text{线性方程}$$

$$\frac{dy}{dx} = x+y$$

$$\frac{dx}{dy} - x = y$$

$$x = e^y \left(C + \int y \cdot e^{-y} dy \right)$$

$$\text{Eq: } f(x) = \sin x - \int_0^x f(u) \cos u du \quad \text{求特解 } f(0)=0$$

$$\text{令 } u = x-t$$

$$\int_0^x f(u) du = \int_0^x f(u) \cos u du$$

$$f(x) = \cos x - f(x)$$

$$\Rightarrow y' = \cos x - y$$

$$\Rightarrow y' + y = \cos x$$

可降阶微分方程

1. $y^{(n)} = f(x)$ 型的微分方程 (不含 y)

令 $z = y^{(n-1)}$ 则 $\frac{dz}{dx} = y^{(n)} = f(x)$

$$z = \int f(x) dx + C_1$$

$$\therefore y^{(n-1)} = \int f(x) dx + C_1$$

$$y^{(n-2)} = \int (\int f(x) dx + C_1) dx + C_2$$

$\vdots$

以此类推

2. $y'' = f(x, y')$ 型的微分方程 (不含 y)

令 $y' = p(x)$ 则 $y'' = p'(x)$

$$p'(x) = f(x, p)$$

$$p(x) = \varphi(x, C_1)$$

$$\frac{dy}{dx} = \varphi(x, C_1)$$

$$dy = \varphi(x, C_1) dx$$

$$y = \int \varphi(x, C_1) dx$$

注: $y^{(n)} = f(x, y^{(n-1)}, \dots, y^{(1)})$

Eg: $y(1+x^2)y'' = 2xy' \rightarrow$ 不含 y 可降阶

$$y|_{x=0}=1, y'|_{x=0}=3$$

令 $y' = p(x)$ 则 $y'' = p'(x)$

$$(1+x^2)p'(x) = 2xp(x)$$

$$\frac{dp(x)}{p(x)} = \frac{2x}{1+x^2} dx$$

$$\int \frac{dp(x)}{p(x)} = \int \frac{2x}{1+x^2} dx$$

$$\ln|p(x)| = \ln|1+x^2| + \ln|C_1|$$

$$p(x) = C_1(1+x^2+1)$$

$$\therefore y'|_{x=0}=3$$

$$\therefore C_1=3$$

$$\therefore y' = 3(1+x^2+1)$$

$$\therefore y = \int 3(1+x^2+1) dx$$

$$\therefore y = x^3 + 3x + C_2$$

$$\therefore y|_{x=0}=1$$

$$\therefore C_2=1$$

$$\therefore y = x^3 + 3x + 1$$

3. $y'' = f(y, y')$ 型的微分方程 (不含 x)

↓
变量

令 $\frac{dy}{dx} = p(y)$ 则 $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} p(y) = \frac{dp(y)}{dy} \cdot \frac{dy}{dx} = p(y) \cdot \frac{dp(y)}{dy}$

$$p \frac{dp}{dy} = f(y, p)$$

$$p = \varphi(y, C_1)$$

$$y' = \varphi(y, C_1)$$

Eg: $y''y - y'^2 = 0$

令 $y' = p(y)$ 则 $y'' = p(y) \cdot p'(y)$

$$y p(y) \frac{dp(y)}{dy} - p^2(y) = 0$$

$$\frac{dp(y)}{dy} = \frac{p(y)}{y}$$

$$\ln|p(y)| = \ln|y| + \ln|C_1|$$

$$p(y) = C_1 y$$

Eg: $y'' - y'^2 = 0$

$$y|_{x=0}=0, y'|_{x=0}=1$$

令 $p(y) = y'$ 则 $y'' = p(y) \frac{dp(y)}{dy}$

$$p(y) \cdot \frac{dp(y)}{dy} = y'^2$$

$$p(y) dp(y) = y'^2 dy$$

$$\frac{1}{2} p^2(y) = \frac{1}{2} y'^2 + C_1$$

$$(y')^2 = y'^2 + 2C_1$$

$$y'|_{x=0}=1, y|_{x=0}=0$$

$$\therefore 1 = 1 + 2C_1 \Rightarrow 2C_1 = 0$$

$$\therefore (y')^2 = y'^2$$

$$\therefore y' = e^y$$

$$\therefore \frac{dy}{e^y} = dx$$

$$\therefore \int \frac{dy}{e^y} = \int dx$$

$$\therefore -e^{-y} = x + C_2$$

$$\therefore y|_{x=0}=0$$

$$\therefore C_2 = -1$$

$$\therefore -e^{-y} = x - 1$$

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一、二阶线性微分方程

$$y'' + p(x)y' + q(x)y = f(x)$$

n 阶线性微分方程的一般形式:

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = f(x)$$

二、线性齐次方程解的结构

定理1: 若函数 y_1, y_2, y_3 是二阶线性方程 $y'' + p(x)y' + q(x)y = 0$

叠0定理 的两个解, 则 $y = C_1 y_1 + C_2 y_2$ 也是该方程的解

$y_1(x), y_2(x)$ 线性无关, 则 $y = C_1 y_1 + C_2 y_2$ 为通解.

可微函数线性相关 $\Leftrightarrow \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} \neq 0$

定理3: 设 $y^{(n)}(x)$ 是二阶非齐次方程

$$y'' + p(x)y' + q(x)y = f(x)$$

的一个特解

$Y_1(x)$ 是相应齐次方程的通解, 则 $y = Y_1(x) + y^{(n)}$ 是非齐次方程的通解

定理4: 设 $y_k(x) (k=1, 2, \dots, n)$ 分别是方程 $y'' + p(x)y' + q(x)y = f_k(x) (k=1, 2, \dots, n)$

的特解, 则 $y = \sum_{k=1}^n y_k$ 是方程 $y'' + p(x)y' + q(x)y = \sum_{k=1}^n f_k$

的特解.

定理3, 定理4 推广至 n 阶非齐次方程 得定理5

三、常数变易法

$y'' + p(x)y' + q(x)y = f(x)$ ①

1. 已知对应齐次方程通解: $y = C_1 y_1(x) + C_2 y_2(x)$

设①的解: $y = y_1(x)u_1(x) + y_2(x)u_2(x)$ 代入①式中

$y' = y_1' u_1 + y_1 u_1' + y_2' u_2 + y_2 u_2'$

$y_1 u_1' + y_2 u_2' = 0$

$y'' = y_1'' u_1 + y_1' u_1' + y_1 u_1'' + y_2'' u_2 + y_2' u_2' + y_2 u_2''$

$y_1'' u_1 + y_1' u_1' + y_2'' u_2 + y_2' u_2' + \underbrace{p y_1 u_1 + p y_2 u_2} + \underbrace{q y_1 u_1 + q y_2 u_2} = f(x)$

$$\begin{cases} y_1 u_1' + y_2 u_2' = 0 \\ y_1' u_1' + y_2' u_2' = f(x) \end{cases} \Rightarrow u_1', u_2' \Rightarrow u_1, u_2 \Rightarrow \text{特解 } y = u_1 y_1 + u_2 y_2$$

2. 仅知①的齐次方程两个非零特解 y_1, y_2

求该齐次方程另一线性无关特解的方法——降阶法

令 $y_3 = u(x)y_1$ (y_1, y_3 线性无关)

代入 $y'' + p(x)y' + q(x)y = 0$

$\Rightarrow u'' y_1 + u' (2 y_1' + p y_1) = 0$

令 u 可得 $u = \text{---}$

常数线性微分方程

1. 二阶常系数线性齐次微分方程: $y'' + p y' + q y = 0$ (p, q 为常数)

令 $y = e^{rx}$ 代入方程中: $r^2 e^{rx} + p r e^{rx} + q e^{rx} = 0 \rightarrow$

特征方程 $r^2 + p r + q = 0$ 可得 r

① $\Delta > 0 \Rightarrow y_1 = e^{r_1 x}, y_2 = e^{r_2 x}$

② $\Delta = 0 \Rightarrow r_1 = r_2 = -\frac{p}{2}$ 用降阶法
 $y_1 = e^{r_1 x}, y_2 = x e^{r_1 x}$

③ $\Delta < 0 \Rightarrow r_{1,2} = \alpha \pm \beta i$
 $y_1 = e^{\alpha x + \beta i x} = e^{\alpha x} (\cos \beta x + i \sin \beta x), y_2 = e^{\alpha x - \beta i x} = e^{\alpha x} (\cos \beta x - i \sin \beta x)$
 $y_1 = \frac{y_1 + y_2}{2} = e^{\alpha x} \cos \beta x, y_2 = \frac{y_1 - y_2}{2i} = e^{\alpha x} \sin \beta x$
 $\frac{y_1}{y_2} = \cot \beta x \neq C$

$$\begin{cases} \Delta > 0 & y = C_1 e^{r_1 x} + C_2 e^{r_2 x} \\ \Delta = 0 & y = C(C_1 + C_2 x) e^{-\frac{p}{2} x} \\ \Delta < 0 & y = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x) \end{cases}$$

推广至 $y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = 0$ (a_k 均为常数)

特征方程: $r^n + a_1 r^{n-1} + \dots + a_{n-1} r + a_n = 0$

若 r 是 k 重实根 则含有 $(C_1 + C_2 x + \dots + C_k x^{k-1}) e^{rx}$

若 $r_{1,2} = \alpha \pm \beta i$ 为重根 则必有 $e^{\alpha x} [(C_1 + C_2 x + \dots + C_k x^{k-1}) \cos \beta x + (D_1 + D_2 x + \dots + D_k x^{k-1}) \sin \beta x]$

2. 二阶常系数线性非齐次微分方程: $y'' + py' + qy = f(x)$

解: $y = \overset{\substack{\uparrow \\ \text{齐次通解}}}{\gamma} + \overset{\substack{\uparrow \\ \text{非齐次特解}}}{y^*}$

待定系数法

① $f(x) = e^{\lambda x} P_m(x)$ 型 (λ 为实数, $P_m(x)$ 为 m 次多项式)

设特解为 $y^* = e^{\lambda x} Q(x)$

$$Q''(x) + (2\lambda + p)Q'(x) + (\lambda^2 + p\lambda + q)Q(x) = P_m(x)$$

1° 若 $\lambda^2 + p\lambda + q \neq 0$ 时 即 λ 不是特征根, $Q(x)$ 为 m 次 $y^* = e^{\lambda x} Q_m(x)$

2° 若 $\lambda^2 + p\lambda + q = 0$ 且 $2\lambda + p \neq 0$ 即 λ 是特征单根, $Q(x)$ 为 $(m+1)$ 次 $y^* = x e^{\lambda x} Q_{m+1}(x)$

3° 若 $\lambda^2 + p\lambda + q = 0$ 且 $2\lambda + p = 0$ 即 λ 是二重特征根, $Q(x)$ 为 $(m+2)$ 次, $y^* = x^2 e^{\lambda x} Q_{m+2}(x)$

若 λ 是 k 重特征根, $y^* = x^k e^{\lambda x} Q_m(x)$
 $k=0, 1, 2$

② $f(x) = e^{\lambda x} [P_l(x) \cos nx + \tilde{P}_l(x) \sin nx]$ 型

利用欧拉公式对 $f(x)$ 变形

$$f(x) = e^{\lambda x} \left[P_l(x) \frac{e^{inx} + e^{-inx}}{2} + \tilde{P}_l(x) \frac{e^{inx} - e^{-inx}}{2i} \right]$$

$$= e^{(\lambda + i n)x} \left[\frac{P_l(x)}{2} + \frac{\tilde{P}_l(x)}{2i} \right] + e^{(\lambda - i n)x} \left[\frac{P_l(x)}{2} - \frac{\tilde{P}_l(x)}{2i} \right]$$

记 $m = \max\{l, n\}$, $f(x) = P_{m+1}(x) e^{(\lambda + i n)x} + \overline{P_{m+1}(x)} e^{(\lambda - i n)x}$

$$= P_{m+1}(x) e^{(\lambda + i n)x} + \overline{P_{m+1}(x)} e^{(\lambda - i n)x}$$

分别求特解

$$\begin{cases} y'' + py' + qy = P_{m+1}(x) e^{(\lambda + i n)x} \\ y'' + py' + qy = \overline{P_{m+1}(x)} e^{(\lambda - i n)x} \end{cases}$$

$$\text{设 } \lambda + i n \text{ 是 } k \text{ 重特征根, } k=0, 1.$$

$$y_1^* = x^k Q_{m+1-k}(x) e^{(\lambda + i n)x}$$

$$y_2^* = \overline{y_1^*}$$

$$\text{求原方程的特解 } y^* = y_1^* + y_2^*$$

$$= x^k e^{\lambda x} [Q_m e^{inx} + \overline{Q_m} e^{-inx}]$$

$$= x^k e^{\lambda x} [Q_m \cos nx + \tilde{Q}_m \sin nx + \overline{Q_m} \cos nx - \tilde{Q}_m \sin nx]$$

$$= x^k e^{\lambda x} [R_{m+1}(x) \cos nx + \tilde{R}_{m+1}(x) \sin nx]$$

欧拉方程

$$x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + \dots + a_n y = f(x)$$

令 $x = e^t$ 则 $t = \ln x$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt}$$

$$\therefore x \frac{dy}{dx} = \frac{dy}{dt}$$

$$\frac{d^2 y}{dx^2} = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d}{dt} \left(\frac{dy}{dt} \right)$$

$$= \frac{1}{x} \frac{d^2 y}{dt^2}$$

$$= \frac{1}{x^2} \frac{d^2 y}{dt^2}$$

$$\therefore x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2} - \frac{dy}{dt}$$

$$\text{令 } D = \frac{d}{dt} \quad Dy = \frac{dy}{dt} \quad D^2 y = \frac{d^2 y}{dt^2}$$

$$xy' = Dy$$

$$x^2 y'' = D(D-1)y$$

$$\vdots$$

$$x^n y^{(n)} = D(D-1)\dots(D-n+1)y \quad (n \leq n)$$